

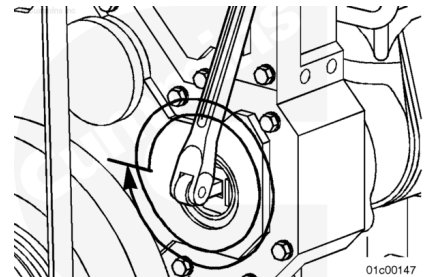
Trainings ▾

Rotation Check

Use **only** the compressor drive or barring device to rotate the crankshaft.

Rotate the crankshaft **clockwise** through two complete revolutions.

If the engine does **not** turn freely, the equipment can have a malfunction or was assembled incorrectly. See equipment manufacturer service information.



Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

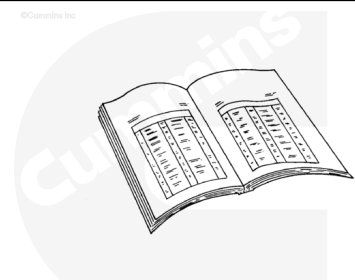
Coolant is toxic. Keep away from children or pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

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Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

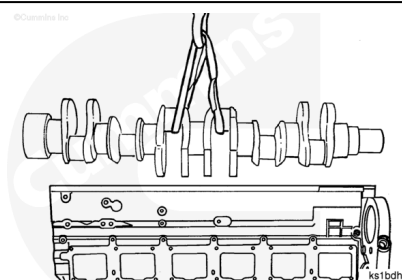
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>)
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>)
- Remove the engine. Refer to Procedure 000-001 in Section 0. (</qs3/pubsys2/xml/en/procedures/10/10-000-001-tr.html>)
- Remove the cylinder heads. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html>)
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/10/10-016-005-tr.html>)
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16. (</qs3/pubsys2/xml/en/procedures/10/10-016-006-tr.html>)
- Remove the front gear housing. Refer to Procedure 001-033 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-033-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html>)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html>)
- Remove the lubricating oil pump. Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html>)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html>)

- Remove the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)
- Remove the main bearing caps. Refer to Procedure 001-006 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-006-tr.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a lifting strap that will **not** damage the crankshaft.

Use a hoist and a sling to remove the crankshaft.

Clean and Inspect for Reuse

Note : Criteria in this procedure applies to crankshafts removed from the engine for rebuild. For in-engine progressive damage inspections of the crankshaft due to debris contamination of the lubricating oil system, refer to Service Bulletin, Crankshaft Reuse Guidelines, Bulletin 5411180.

Use crocus cloth or a Scotch-Brite™ pad to remove discoloration or light scratches from the machined surfaces of the crankshaft.

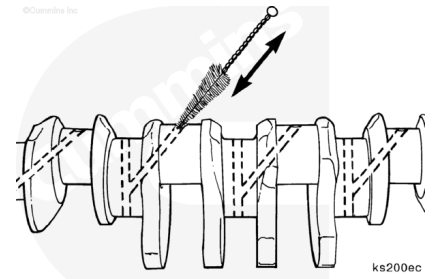


⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

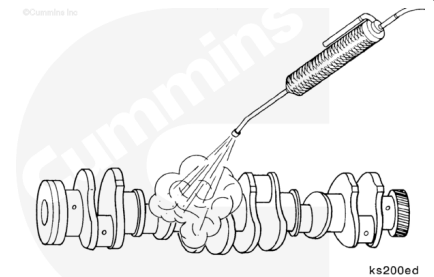
possibility of personal injury.

Use a bristle brush and solvent to clean all of the crankshaft oil drillings.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.



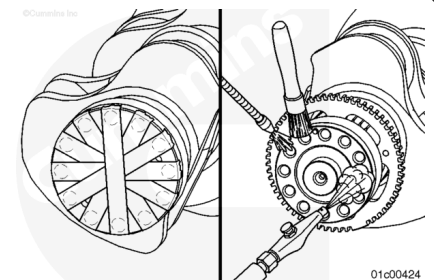
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean the crankshaft and dry with compressed air.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not use a thread chaser to clean the capscrew threads in the crankshaft. Severe engine damage can result.

ISX applications use rolled capscrew threads in the capscrew holes of the crankshaft.

To clean the rolled threads, flush with solvent, and dry with compressed air.

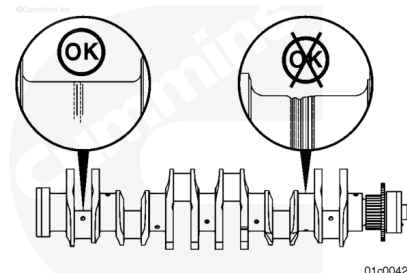
If additional cleaning is required, brush with a nylon bristle brush.

Place tape over the threaded capscrew holes.

Inspect the machined surfaces for scratches or nicks.

Use a crocus cloth to remove any nicks and scratches.

If scratches or nicks can be felt with a fingernail after the crankshaft has been polished with crocus cloth, the crankshaft **must** be replaced.



Inspect the crankshaft for dynamic balancing marks. One or more of the crankshaft counterweights will contain a drilled hole or drilled start point indicating that the crankshaft has been balanced.

Note : This applies only to engines built March 2009 or later.

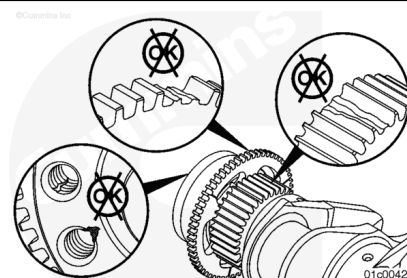
Inspect the threaded capscrew holes for damage.

If the threaded capscrew holes are damaged, the crankshaft **must** be replaced.

Inspect the crankshaft gear and tone wheel for excessive wear or damage.

If any teeth on the crankshaft gear are missing, chipped, or pitted, the crankshaft gear **must** be replaced.

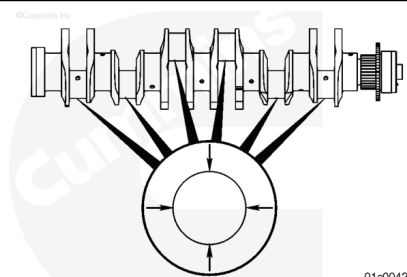
If any teeth on the tone wheel are missing, the tone wheel **must** be replaced.



Measure the crankshaft connecting rod journal outside diameter.

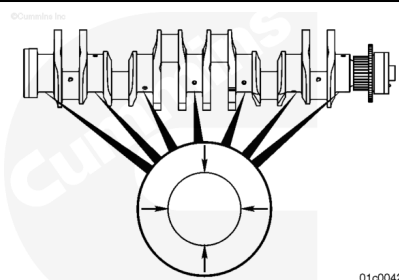
Crankshaft Connecting Rod Journal Outside Diameter		
mm		in
93.98	MIN	3.700
94.02	MAX	3.702

If the crankshaft connecting rod journal outside diameter is **not** within specifications, the crankshaft **must** be replaced.



Measure the crankshaft main bearing journal outside diameter.

Crankshaft Main Bearing Journal Outside Diameter		
mm		in
126.98	MIN	4.999
127.02	MAX	5.001

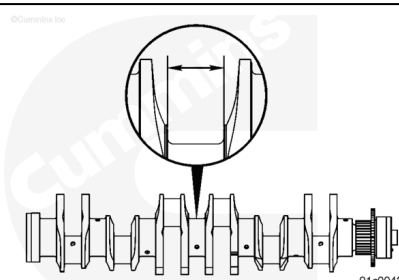


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If the crankshaft main bearing journal outside diameter is **not** within specifications, the crankshaft **must** be replaced.

Measure the crankshaft thrust face width.

Crankshaft Thrust Face Width		
mm		in
54.95	MIN	2.163
55.05	MAX	2.167

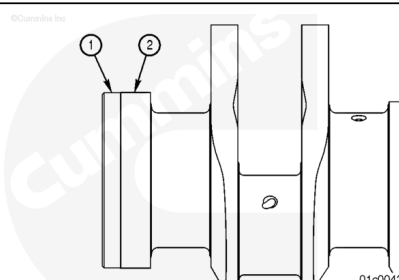


01c00429

If the crankshaft thrust face width is **not** within specifications, the crankshaft **must** be replaced.

Measure the crankshaft rear oil seal flange outside diameter.

Crankshaft Rear Oil Seal Flange Outside Diameter			
	mm		in
Flange Diameter Location (1)	164.964	MIN	6.495
	165.035	MAX	6.497
Flange Diameter Location (2)	165.475	MIN	6.515
	165.525	MAX	6.517



01c00430

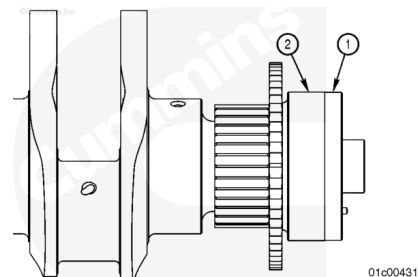
If the crankshaft rear oil seal flange outside diameter is **not** within specifications, the crankshaft **must** be replaced.

Check the crankshaft rear oil seal flange for nicks, burrs, and grooves.

If a fingernail catches in a nick, burr, or groove, the crankshaft **must** be replaced. Damage to the sealing surface can result in a seal leak.

Measure the crankshaft front oil seal flange outside diameter.

Crankshaft Front Oil Seal Flange Outside			
	mm		in
Flange Diameter Location (1)	139.24	MIN	5.482
	139.63	MAX	5.497
Flange Diameter Location (2)	139.89	MIN	5.507
	140.04	MAX	5.513



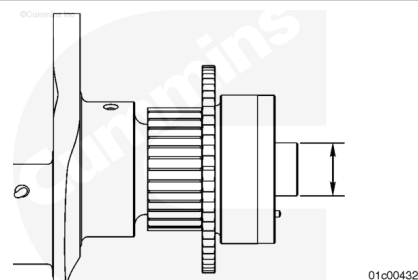
If the crankshaft front oil seal flange outside diameter is **not** within specifications, the crankshaft **must** be replaced.

Check the crankshaft front oil seal flange for nicks, burrs, and grooves.

If a fingernail catches in a nick, burr, or groove, the crankshaft **must** be replaced. Damage to the sealing surface can result in a seal leak.

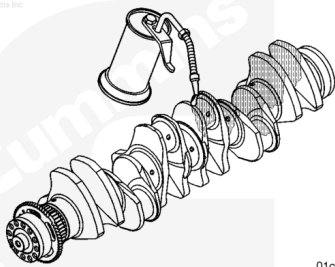
Measure the crankshaft damper pilot outside diameter.

Crankshaft Pilot Outside Diameter		
mm		in
49.985	MIN	1.968
50.015	MAX	1.969



If the crankshaft pilot outside diameter is **not** within specifications, the crankshaft **must** be replaced.

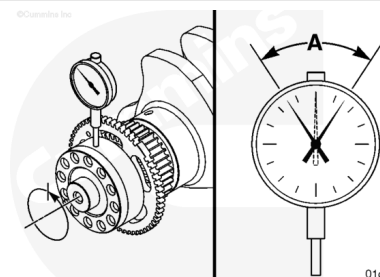
Lubricate the entire crankshaft with clean 15W-40 oil to prevent rust.



01c00433

Bend and Twist Inspect

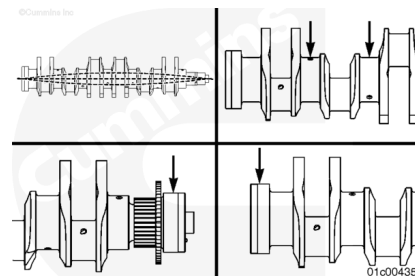
The crankshaft alignment limits are based on total indicator runout. The runout is measured as the crankshaft is turned one revolution while recording the total movement of the dial indicator.



01c00434

When measuring the crankshaft alignment, four measurements **must** be taken:

- Full length alignment
- Bearing to bearing runout
- Rear oil seal flange diameter runout
- Front oil seal crankshaft adapter diameter runout.



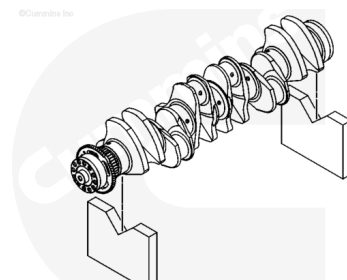
01c00435

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

Measuring from the top of the main bearing journal will cause inaccurate alignment measurements because of crankshaft sag.

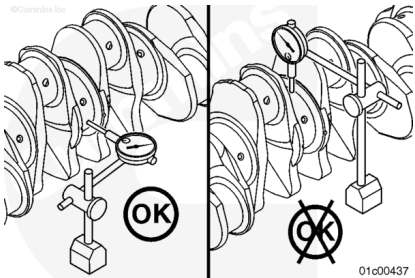


01c00436

To check the crankshaft full length alignment, support the crankshaft on the number one and number seven main bearing journals.

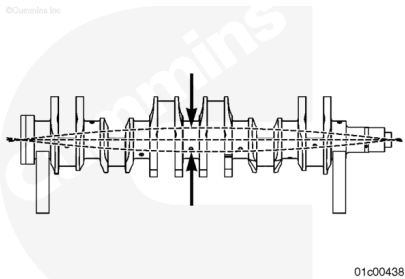
The dial indicator **must** be set up so the indicator is located on the side of the number four main bearing journal.

Set up the dial indicator.



Rotate the crankshaft one complete revolution and record the total indicator runout.

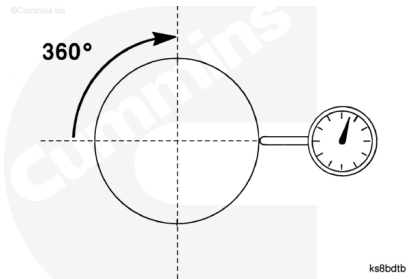
Total Indicator Runout		
mm		in
0.200	MAX	0.008



If the crankshaft total indicator runout is **not** within specifications, the crankshaft **must** be replaced.

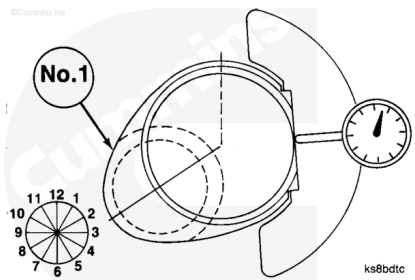
Bearing to bearing runout is defined as the relationship of the total indicator runout of a main bearing journal as it is rotated on a common axis to the total indicator runout of an adjacent journal.

Bearing to bearing runout is often referred to as step runout, adjacent runout or journal to journal runout.



The clock position is defined as the location of the journal at the highest total indicator runout point. Compare its angular relationship with the number one crankshaft pin, as viewed from the front of the crankshaft.

In the illustration, the crankshaft pin is at the 8 o'clock position. This is the clock position of the journal being measured.



⚠ WARNING ⚠

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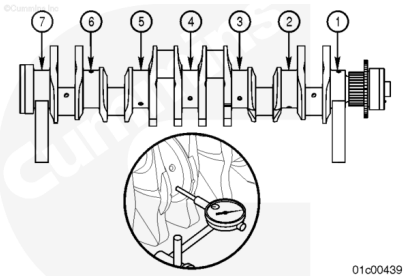
component or assembly.

⚠ CAUTION ⚠

Measuring from the top of the main bearing journal will cause inaccurate alignment measurements because of crankshaft sag.

To check the crankshaft full length alignment, support the crankshaft on the number one and number seven main bearing journals.

Set the up dial indicator on the side of the main bearing journal.



Rotate the crankshaft and measure the total indicator runout at each bearing journal.

Record the value and clock position for each location.

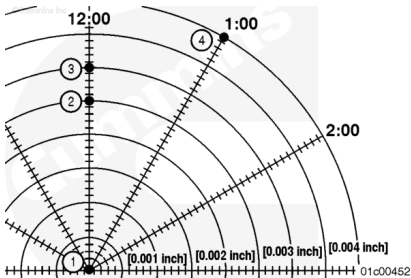
ITEM	RUNOUT TIR [INCH]	CLOCK POSITION
JOURNAL STEP		
1	[0]	0
2	[0.0021]	12
3	[0.0030]	12
4	[0.0039]	1
5	[0.0025]	1
6	[0.0016]	2
7	[0]	0

ks8bdtf

For each journal, plot the total indicator runout value at its clock position on a polar chart.

The end journals, supported by vee blocks, **must** be plotted at the center of the chart.

The illustration illustrates the plot points.



Journal	Total Indicator Runout	Clock Position
(1)	0	0
(2)	0.002	12
(3)	0.003	12
(4)	0.004	1

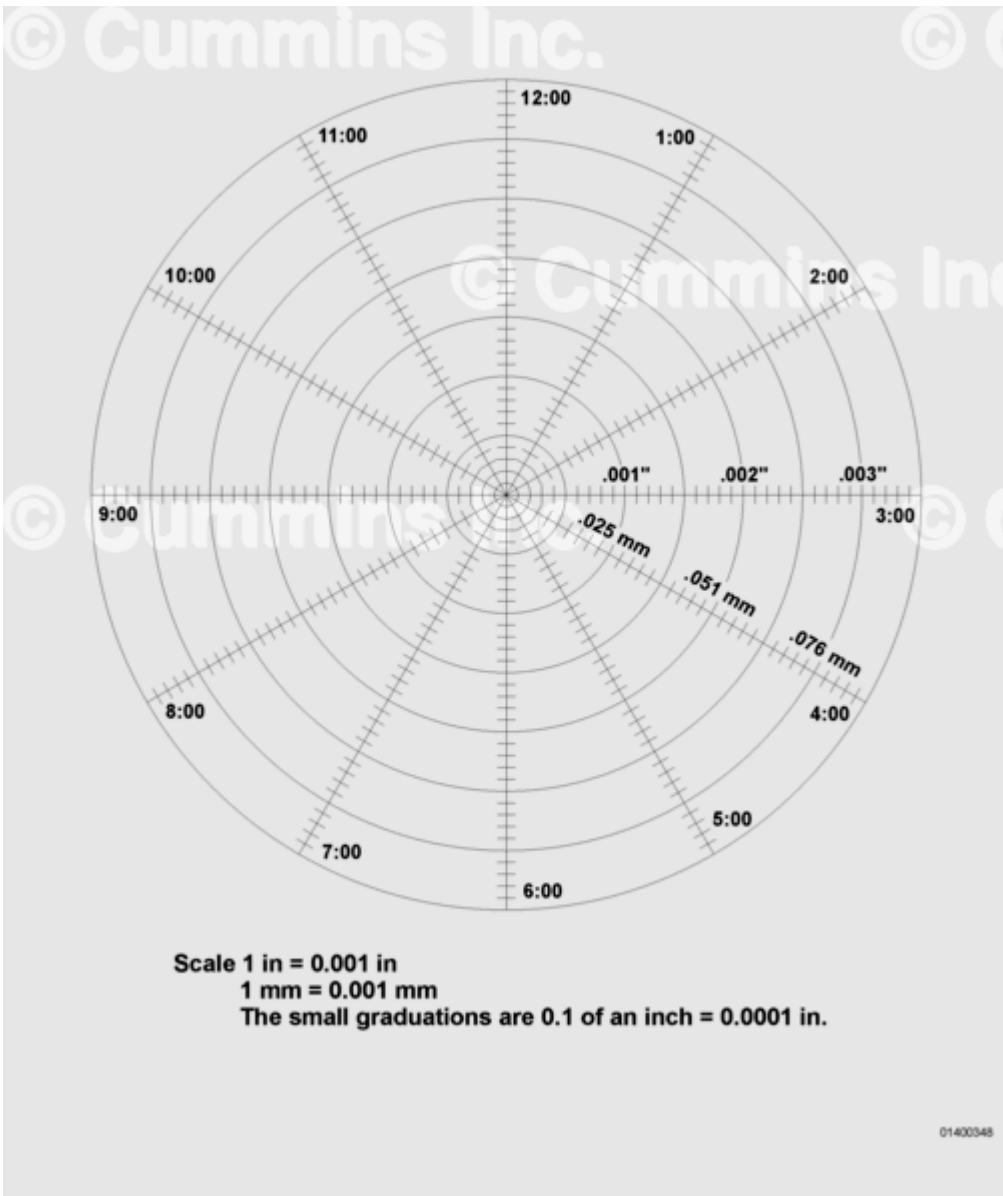
Draw a straight line between the plotted points. From journal number one, to journal number two, to journal number three, until all journals are plotted on the chart.

To determine the adjacent journal runout, measure the length of the line from each journal to its corresponding journal point.

In the above table journal number three and number four are 51 mm [2 in]. This represents a runout of 0.051 mm [0.002 in].

Record the adjacent runout for each main bearing journal.

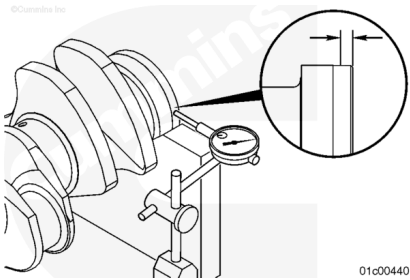
Adjacent Journal Runout		
mm		in
0.075	MAX	0.003



If the alignment between adjacent journals is **not** within specifications, it can **not** be repaired by grinding. The crankshaft **must** be replaced. ISX crankshafts are fully fillet hardened and can **not** be straightened.

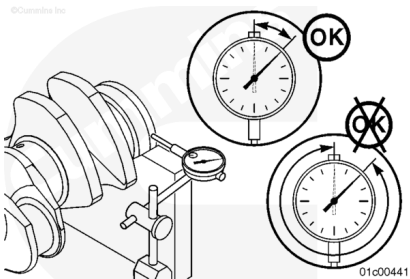
Rear oil seal flange diameter runout measurement:

With the crankshaft supported on the number one and number seven main bearing journals, position a dial indicator 25.4 mm [1.0 in] from the flywheel mounting face of the crankshaft.



Rotate the crankshaft one complete revolution and record the total indicator runout.

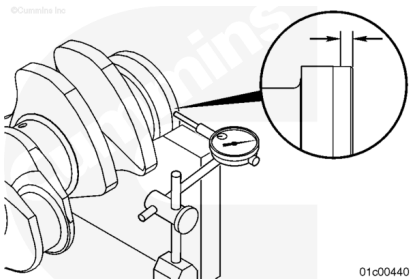
Rear Oil Seal Flange Total Indicator Runout		
mm		in
0.03	MAX	0.0012



If the rear oil seal flange total indicator runout is **not** within specifications, the crankshaft **must** be replaced.

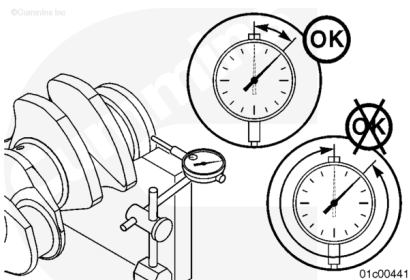
Front oil seal crankshaft adapter diameter runout measurement:

With the crankshaft supported on the number one and number seven main bearing journals, position a dial indicator 25.4 mm [1.0 in] from the balancer mounting face of the crankshaft.



Rotate the crankshaft one complete revolution and record the total indicator runout.

Front Oil Seal Crankshaft Adapter Diameter Runout		
mm		in
0.05	MAX	0.002

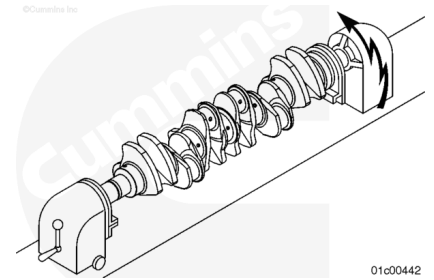


If the front oil seal crankshaft adapter diameter runout is **not** within specifications, the crankshaft **must** be replaced.

Magnetic Crack Inspect

The crankshaft **must** be tested by the "continuous method". The entire surface **must** be wet with the magnetic bath solution before and during the flow of magnetic current through the crankshaft.

For the head shot (longitudinal magnetization) method, apply the magnetic bath solution to the crankshaft and use 3800 amperes with VDC or rectified VAC to magnetize the crankshaft.



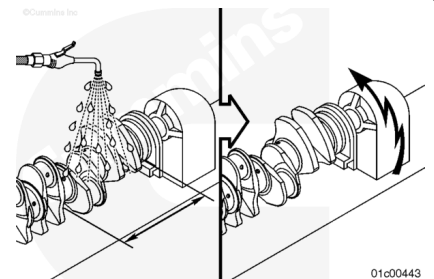
A minimum of three successive head shots are required for each section of the crankshaft.

Flow the magnetic solution over the first 1/3-length of the crankshaft and apply two head shots of magnetizing current.

To prevent washing the metal particles off the crankshaft, turn the magnetic solution flow off.

Apply a third head shot of magnetizing current.

Inspect the crankshaft for open indications.

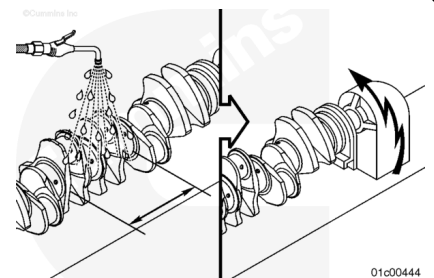


Flow the magnetic solution over the second 1/3-length of the crankshaft and apply two head shots of magnetizing current.

To prevent washing the metal particles off the crankshaft, turn the magnetic solution flow off.

Apply a third head shot of magnetizing current.

Inspect the crankshaft for open indications.

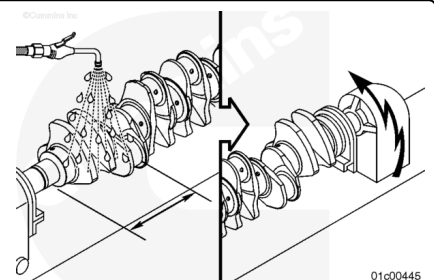


Flow the magnetic solution over the last 1/3-length of the crankshaft and apply two head shots of magnetizing current.

To prevent washing the metal particles off the crankshaft, turn the magnetic solution flow off.

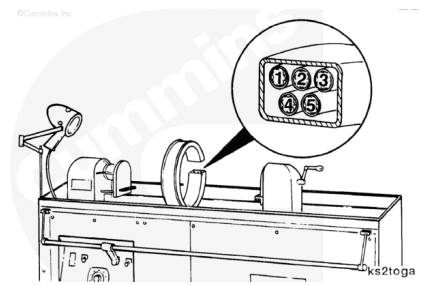
Apply a third shot of magnetizing current.

Inspect the crankshaft for open indications.



For the coil shot (circumferential magnetization) method, apply the magnetic solution and magnetize the crankshaft in a 51.35 mm [20.249 in] inside coil or equivalent. Use 3,600 to 4,000 ampere-turns with VDC or rectified VAC.

Ampere-turn is the amperage flowing through the coil multiplied by the number of turns in the coil.



A minimum of three coil shots are required for each 457 mm [18 in] length to correctly magnetize the crankshaft.

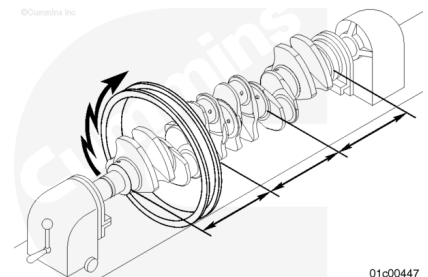
Flow the magnetic solution over the crankshaft in 1/3 sections.

Apply two coil shots, after wetting with the magnetic solution, with the coil placed at $\frac{1}{4}$ -length points along the crankshaft.

To prevent washing the metal particles off the crankshaft, turn the magnetic solution flow off.

Apply the third coil shot with the coil placed at $\frac{1}{4}$ -length points along the crankshaft.

Inspect each 457 mm [18 in] section after the third coil shot is completed.

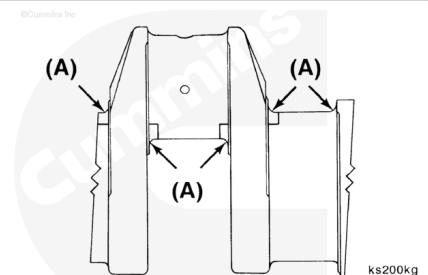


An open indication is a crack open to the surface that can be observed without using optical enhancement.

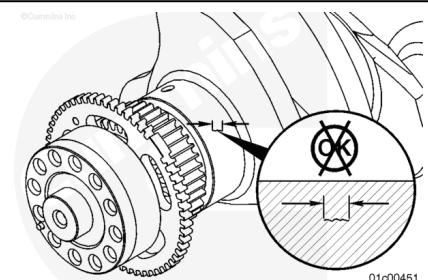
An open indication can sometimes be felt with a fingernail.

Do **not** mistake forging trim lines for crack indications.

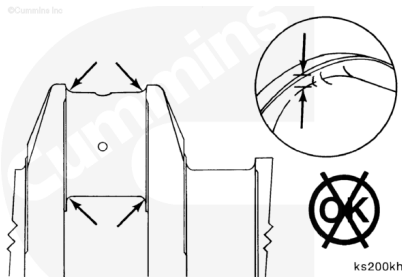
If open indications are observed in the hardened fillet area or the critical areas shown (A), the crankshaft **must** be replaced.



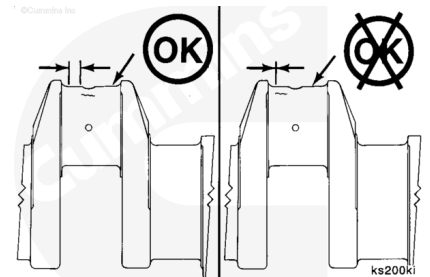
Open indications longer than 6 mm [0.25 in] are **not** acceptable.



Open indications in any fillets or the critical areas are **not** acceptable.

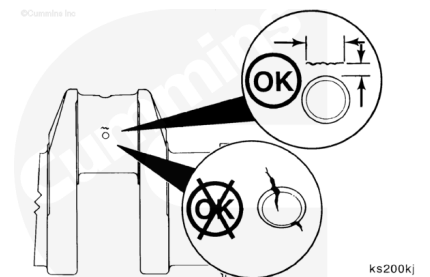


Open indications on the crankpin or main bearing journal surfaces that extend closer than 3 mm [0.12 in] to the fillet, but do **not** enter the fillet, are acceptable.



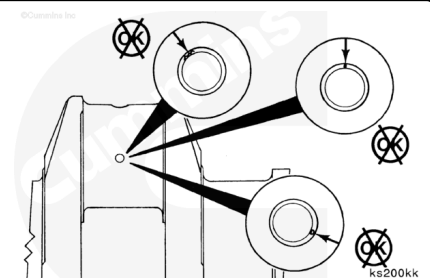
Open indications in the longitudinal direction up to 12.5 mm [0.5 in] in length, but **not** closer than 3 mm [0.12 in] to the oil hole chamfer, are acceptable.

Open indications that enter the oil hole chamfer or intersect the oil hole at 45 degrees $\pm$ 10 degrees are **not** acceptable.

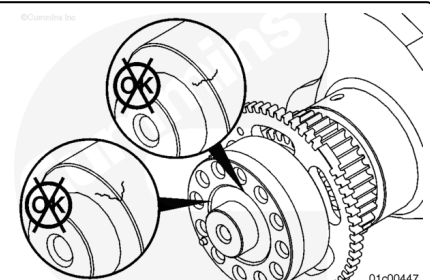


Any open indication in the crankpin bearing hole entrance chamfer is **not** acceptable.

Open indications in the counterweights inside the critical area are **not** acceptable.

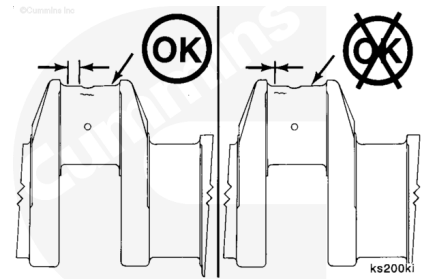


Open indications that extend over the face of the crankshaft or extend into the damper mounting capscrew holes are **not** acceptable.



Subsurface indications can **not** be seen with the eye after magnetic particles have been removed. These are indications of open areas or foreign objects below the surface.

Inspect for subsurface indications.



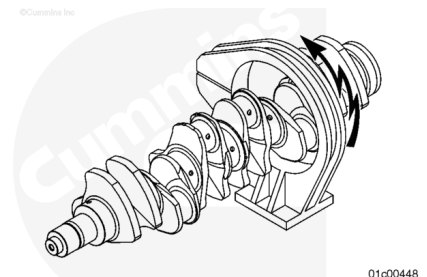
⚠ CAUTION ⚠

All magnetism must be removed from the crankshaft completely and the crankshaft cleaned thoroughly. Small metal particles will cause engine damage.

Demagnetize the crankshaft.

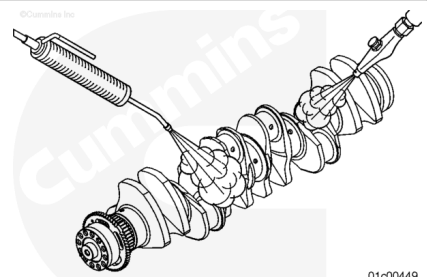
The maximum residual magnetic field on any pin or main bearing journal is 8 gauss.

It will possibly be necessary to use reverse D.C.-30 point step down equipment, or equivalent, to remove the magnetism.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.



⚠ WARNING ⚠

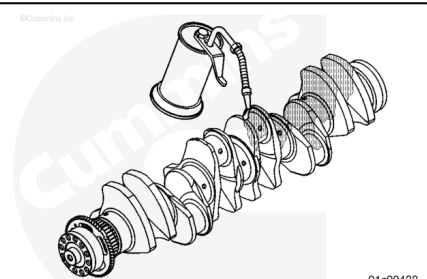
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean the crankshaft.

Dry the crankshaft with compressed air.

Use a light preservative oil to lubricate the crankshaft to prevent rust.

If the crankshaft is **not** going to be used immediately, protect it with a plastic cover to prevent dirt sticking to the oil.



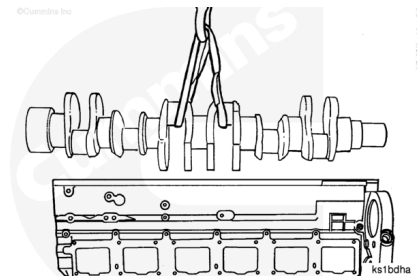
Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

Use a lifting strap that will not damage the crankshaft. Do not drop the crankshaft on the bearings.



Install the main bearings. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-006-tr.html>)

The end of the crankshaft with the smallest diameter **must** point toward the front of the cylinder block.

Install the crankshaft.

Finishing Steps

- Install the main bearing caps. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-006-tr.html>)
- Install the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html>)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html>)
- Install the lubricating oil pump. Refer to Procedure 007-031 in Section 7.



(/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)

- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Install the front gear housing. Refer to Procedure 001-033 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-033-tr.html)
- Install the flywheel housing. Refer to Procedure 016-006 in Section 16. (/qs3/pubsys2/xml/en/procedures/10/10-016-006-tr.html)
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/10/10-016-005-tr.html)
- Install the cylinder heads. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
- Install the engine. Refer to Procedure 000-002 in Section 0. (/qs3/pubsys2/xml/en/procedures/10/10-000-002-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)